

TECHNICAL REQUIREMENTS DOCUMENT

AskBeforeAnswer — Clarify-or-Answer Decision Model

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1. Purpose & Scope

This Technical Requirements Document (TRD) specifies the functional and non-functional requirements for AskBeforeAnswer, a fine-tuned Qwen 2.5 7B language model that decides, for every incoming user question, whether to **ask a targeted clarifying question** or **answer directly**. The document defines the dataset, training, evaluation, reproducibility, experiment-tracking, versioning, and deployment requirements the system must satisfy prior to production promotion.

The scope of this document covers:

- The clarify-or-answer decision objective and its structured 4-field output schema
- Dataset construction, splits, and quality requirements for SFT and DPO/GRPO training
- Minimum model-performance bars and the evaluation methodology used to certify them
- Training reproducibility, configuration discipline, and experiment-tracking requirements
- Model registry promotion, versioning, and Hugging Face Hub release gating
- Deployment and inference-serving requirements for the production Streamlit application

2. Reference Documents & Standards

Reference	Source	Applicability
AmbigNQ Corpus	Sewon Min et al. / AI2	Base ambiguous-question source data
DVC Documentation	dvc.org	Data & pipeline versioning
Hydra Documentation	hydra.cc	Configuration composition and override

Reference	Source	Applicability
Weights & Biases Docs	docs.wandb.ai	Experiment tracking, sweeps, model registry
W&B Weave Docs	weave-docs.wandb.ai	LLM observability, evaluation, tracing
Hugging Face Hub Docs	huggingface.co/docs	Model & dataset publication
DPO Paper (Rafailov et al., 2023)	arXiv:2305.18290	Preference-optimization objective
Unsloth Documentation	unsloth.ai/docs	Accelerated LoRA fine-tuning

3. Objective & Problem Statement

Open-domain question-answering models frequently guess a user's intent when a query is ambiguous (e.g. *"How do I make pasta?"* admits many valid interpretations), which leads to hallucinated or mismatched answers. The **primary training objective** of this project is to align a Qwen 2.5 7B base model to a **clarification-first policy**: for every input question the model must emit a decision, $\text{Action} \in \{\text{Clarify}, \text{Answer}\}$, together with the reasoning and content required to act on that decision.

3.1 Structured Output Contract

Every model response, in both training targets and production inference, must conform to a strict 4-field schema:

Field	Type	Description
Action	Enum {Clarify, Answer}	Binary routing decision for the query
Reasoning	Free text (CoT)	Chain-of-Thought trace explaining why the question is ambiguous or clear
Facets	JSON array of strings	Open-ended semantic attributes missing from the query; empty when Action=Answer
Response	Free text	The targeted clarifying question, or the final direct answer

NOTE Facets are **open-ended**. The generation prompts used to build the training data do not restrict the model to a fixed, hardcoded facet taxonomy — the instructor model dynamically produces the semantic categories relevant to each question.

4. Functional Requirements

ID	Requirement	Priority
FR-1	Given a user question, the model MUST output a single Action of Clarify or Answer.	P0
FR-2	When Action = Clarify, the Facets array MUST be non-empty and the Response MUST contain a targeted clarifying question.	P0

ID	Requirement	Priority
FR-3	When Action = Answer, the Facets array MUST be empty and the Response MUST contain a direct, factual answer.	P0
FR-4	The model output MUST always include a Reasoning field produced before the Action/Response (Chain-of-Thought first).	P0
FR-5	The training pipeline MUST support at least four fine-tuning methods: SFT, DPO, ORPO, and GRPO, selectable as independent training variants.	P0
FR-6	The inference pipeline (ClarifyOrActPipeline) MUST parse the raw generation into the 4-field schema and route the UI response accordingly.	P0
FR-7	Model promotion to production MUST require explicit human review of generated telemetry reports (human-in-the-loop gate).	P0
FR-8	GRPO reward shaping MUST evaluate format adherence, action correctness, facet/action logical consistency, and factual accuracy as independent, re-weightable reward terms.	P1

5. Dataset Requirements

5.1 Source & Construction

Property	Detail
Source corpus	AmbigNQ (ambiguous open-domain QA)
Generator / instructor model	qwen2.5-7b-instruct
Schema	Strict 4-field JSON: Action, Reasoning, Facets, Response
Chosen (y+) synthesis	Instructor model acting as expert agent, producing the SFT-quality gold target
Rejected (y-) synthesis	Prompt-Guided Synthetic Generation — negative system prompt yields structurally perfect but logically flawed reasoning
Symmetry requirement	Chosen/rejected pairs MUST be structurally symmetric so DPO margins isolate reasoning quality, not formatting
Pipeline stages	(1) Schema enforcement & open-ended facet extraction → (2) SFT chosen synthesis → (3) DPO rejected synthesis

5.2 Dataset Quality Requirements

ID	Requirement
DR-1	Every record MUST validate against the 4-field JSON schema before being written to sft_train.jsonl / dpo_train.jsonl.
DR-2	Facets MUST be a parseable JSON array for every Clarify-labeled record and MUST be empty for every Answer-labeled record.

ID	Requirement
DR-3	Rejected samples MUST NOT be produced via expensive model-grounded mining; Prompt-Guided Synthetic Generation is the required (cheaper, native) method.
DR-4	Dataset generation MUST be executed and tracked via DVC (make preprocess) so raw → processed lineage is reproducible.
DR-5	The final dataset MUST be published to the Hugging Face Hub as a versioned dataset repository with an accompanying Data Card.

NOTE Chosen and rejected DPO targets must remain perfectly symmetric in structure. Because DPO optimizes log-probability margins over the entire sequence, asymmetric formatting between chosen and rejected samples would let the model reward-hack on formatting instead of learning correct reasoning.

6. Model Performance Requirements

Six model variants (base, sft, dpo_only, sft_dpo, clarifier_lora, orpo, grpo) are evaluated on the sewon_ambig_qa_eval benchmark. The following minimum bars apply to any variant considered for production promotion under the **balanced policy** (strong clarification behavior without refusing to answer unambiguous questions):

Gate	Threshold	Action on Failure
Macro F1 (Action classification)	≥ 0.55	Block promotion
Answer F1	≥ 0.35	Block promotion; flag mode collapse risk
Clarify F1	≥ 0.70	Block promotion
Ambiguity Detection F1	≥ 0.94	Warning; manual review
Facet Generation Rate	≥ 0.95	Block; indicates schema non-adherence
Clarify Ratio (pred/gold clarify rate)	0.9 – 1.3	Warning; indicates over- or under-clarification

IMPORTANT No single model dominates every metric. A model that refuses to answer unambiguous questions (e.g. the clarifier_lora variant, answer_f1 = 0) suffers from mode collapse and MUST be rejected for general-purpose deployment even if its Clarify F1 is highest.

7. Evaluation Metrics & Methodology

7.1 Dual-Scoring Evaluation

Evaluator	Method	Captures
LLM-as-a-Judge (Gemini 2.5 Flash / Gemma 4)	W&B Weave scorer	Ambiguity Detection F1, Clarification Quality F1, Usefulness
ActionScorer (rule-based)	Deterministic string/JSON comparison	Model Accuracy — raw Action correctness vs. ground truth

7.2 Definitive Model Ranking (Balanced Policy)

Rank	Model	Macro F1	Answer F1	Clarify F1	Assessment
1st	sft	0.615	0.485	0.746	Best calibration; clarify_ratio (1.23) closest to ideal 1.0
2nd	dpo_only	0.553	0.357	0.750	Doubles base answer capability while holding ambiguity detection at 0.964
3rd	grpo	0.532	0.308	0.757	Highest clarify_f1 of any functional model; precise facet extraction

sft_dpo (4th) over-corrects toward caution (answer_f1 = 0.296); base (5th) over-clarifies unambiguous questions; clarifier_lora (6th) exhibits complete mode collapse (answer_f1 = 0) and MUST NOT be promoted.

Run the full evaluation suite and publish the Weave leaderboard with:

```
make evaluate
```

8. Training Requirements

ID	Requirement
TR-1	Training MUST be orchestrated exclusively through DVC pipeline stages defined in dvc.yaml; ad-hoc script invocation outside DVC is prohibited for tracked runs.
TR-2	All hyperparameters and dependencies MUST be resolved through Hydra configuration composition; no hardcoded training constants are permitted in source code.
TR-3	The training stack MUST support Unsloth-accelerated LoRA fine-tuning with Flash Attention (Ampere+) and xFormers fallback (Turing/Volta) for attention computation.
TR-4	When use_unsloth: true is set but the Unsloth package is unavailable (e.g. CPU-only macOS), the trainer MUST gracefully fall back to native Hugging Face model loading.
TR-5	GRPO training MUST expose four independently weighted reward functions (format, action, facet-logic, accuracy) via configs/training/grpo.yaml reward_weights.
TR-6	HPC / Slurm training environments MUST load a CUDA toolkit module compatible with Unsloth's on-the-fly Triton kernel compilation prior to pipeline execution.

ID	Requirement
TR-7	Each training variant (sft, sft-only, dpo-only, orpo, grpo) MUST be runnable both via a generic make train TRAIN_VARIANT=<name> and via a readable alias (e.g. make train-dpo).

9. Reproducibility Requirements

ID	Requirement
RP-1	A full pipeline re-run (dvc repro) from a given dvc.lock + conf/ snapshot MUST reproduce identical metrics within floating-point tolerance.
RP-2	Every model artifact promoted toward production MUST have traceable lineage: DVC experiment ID + Git SHA + Hydra config snapshot + W&B run ID.
RP-3	Large or generated ML artifacts (datasets, checkpoints) MUST NOT be committed to Git; DVC is the system of record for all binary artifacts.
RP-4	Sweep trials MUST be distinguishable from clean baseline runs via a W&B .sweep tag so ablation reports can programmatically exclude sweep noise.
RP-5	Applying a sweep's winning configuration MUST be a single deterministic operation: dvc exp apply sweep_<Run ID>, followed by a Git commit of the updated YAML defaults.

10. Experiment Tracking Requirements

ID	Requirement
ET-1	All training runs MUST stream live metrics (loss curves, reward components, eval scores) to Weights & Biases.
ET-2	Hyperparameter sweeps MUST use W&B Bayesian Sweeps for SFT, DPO, and GRPO, each isolating its own key hyperparameter set (LR/batch/epochs/LoRA rank for SFT; beta/LR for DPO; beta/LR/reward coefficients for GRPO).
ET-3	Post-sweep, scripts/generate_sweep_report.py MUST statelessly regenerate docs/sweep_report.md, including Validation Curves grouped by Sweep ID.
ET-4	All qualitative and quantitative evaluation runs MUST be logged to W&B Weave, producing a dynamic leaderboard and full input/output traces.
ET-5	The production Streamlit application MUST log every user interaction, prompt, and generation to Weave for real-time observability.

11. Model / Versioning Requirements

Hugging Face deployment is permitted **only** from a W&B production artifact whose exact identity and immutable digest have been recorded in release provenance and independently verified at deployment time.

Stage	Command	Gate Performed
1. Promote DVC experiment	make promote-dvc MODEL=<model> EXPERIMENT=<id>	Marks the DVC experiment as the selected source of truth (does not deploy)
2. Publish W&B artifact	make publish-model-artifact MODEL=<model> EXPERIMENT=<id> STAGE=production	Validates model, creates/verifies immutable artifact, checks registry digest + production alias, records provenance
3. Deploy to Hugging Face	make deploy-hf	Reads verified provenance, confirms digest, pushes exact artifact + updated Model/Data Card

12. Deployment / Inference Requirements

ID	Requirement
DI-1	The deployed inference application MUST require only src/ask_before_answer/inference/pipeline.py and app/ — the data, training, and evaluation modules MUST NOT be part of the deployment payload.
DI-2	Deployment MUST be triggered exclusively by a published GitHub Release, executed by .github/workflows/deploy-hf-demo.yml.
DI-3	The Streamlit application MUST load the production model from the Hugging Face model repository at runtime, not from a bundled checkpoint.
DI-4	Minimum inference hardware: 1x NVIDIA T4/L4/RTX 3060 (8–12GB VRAM) at 4-bit/8-bit precision; CPU-only inference is supported but not recommended for production.
DI-5	All user interactions on the deployed application MUST be traced to W&B Weave for observability.

13. Constraints & Acceptance Criteria

Category	Requirement	Specification
Reproducibility	Full pipeline re-run	dvc repro from dvc.lock + conf/ snapshot MUST produce identical metrics within floating-point tolerance
Governance	Promotion gate	No model reaches production without explicit human review of generated telemetry (human-in-the-loop, non-negotiable)
Performance	Minimum viable policy	Promoted model MUST satisfy Section 6 thresholds AND avoid mode collapse (answer_f1 > 0)
CI Scope	CI environment	GitHub Actions CI runs CPU-only lint + tests; GPU training dependencies (CUDA PyTorch, Unsloth, xFormers, Flash Attention) are intentionally excluded from CI
Artifact hygiene	Git repository	No large or generated ML artifacts committed to Git; DVC is authoritative for data/model binaries
Security	Secrets management	No credentials in configuration files; all secrets (HF_TOKEN, GEMINI_API_KEY, WANDB keys) via .env / GitHub Secrets

14. Dependencies & Version Requirements

Package	Min Version	Purpose
Python	3.10 (≥3.9 supported)	Runtime environment
PyTorch	2.6+	Core tensor / training framework (CUDA 12.1+ or 13.0)
transformers	4.4x	Base model loading and generation
unsloth	latest	Accelerated LoRA fine-tuning, custom Triton kernels
xformers	latest	Fallback attention kernels for older GPU architectures
flash-attn	2.x	Memory-efficient attention (Ampere+ GPUs)
bitsandbytes	latest	4-bit / 8-bit QLoRA quantization
dvc	3.x	Data & pipeline versioning, reproducible experiments
hydra-core	1.3	Configuration composition and override
wandb	latest	Experiment tracking, sweeps, model registry
weave	latest	LLM observability and evaluation leaderboards
huggingface_hub	latest	Model / dataset publication
streamlit	latest	Production inference UI
pytest	7.4	Unit and integration tests

15. Open Items & Decisions Required

#	Open Item	Owner	Target Date
1	Finalize minimum-viable thresholds for facet_generation_rate under GRPO reward shaping	ML Research	TBD
2	Decide whether sft_dpo should remain a supported production candidate given its answer_f1 regression	Applied NLP	TBD
3	Confirm long-term storage / retention policy for DVC-tracked model artifacts	MLOps	TBD
4	Evaluate whether a dedicated accelerated serving backend is warranted for the HF Space demo	ML Engineering	TBD
5	Define SLA for Weave-observed production drift and the associated re-training trigger	ML Research	TBD

16. Revision History

Version	Date	Author	Change Summary
1.0.0	Sep 2026	ML Research & Applied NLP	Initial draft for review

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